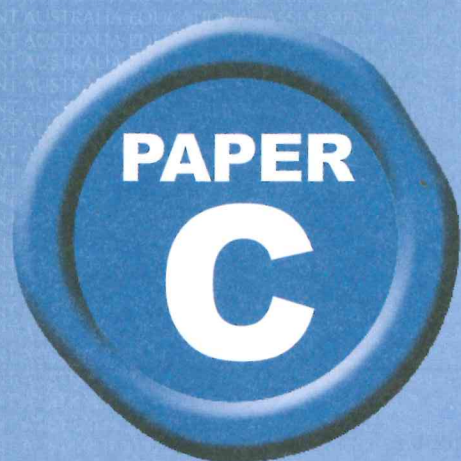




UNSW
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

**EDUCATIONAL ASSESSMENT
AUSTRALIA**

Year 5



ICAS

**INTERNATIONAL COMPETITIONS AND
ASSESSMENTS FOR SCHOOLS**

SCIENCE

2009

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 55 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

**PLEASE SEE BACK COVER FOR A LIST
OF THE YEAR LEVELS THAT SHOULD
SIT THIS PAPER**

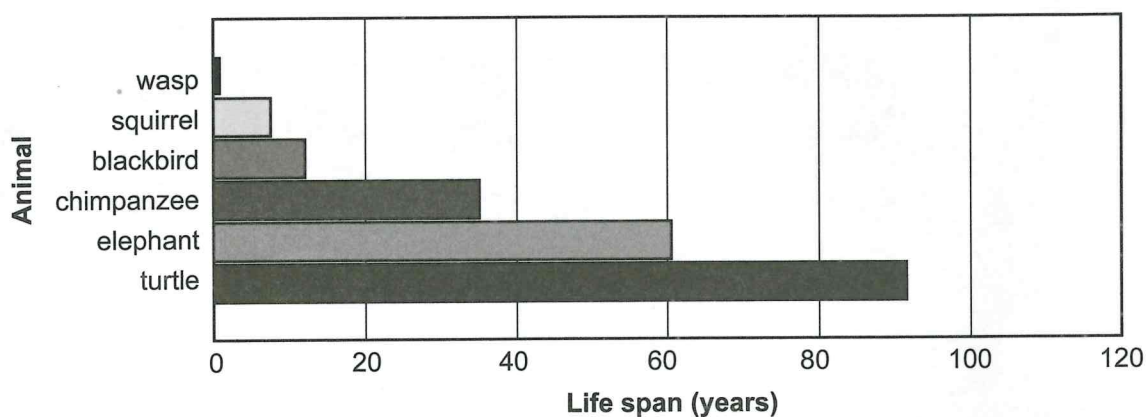
1. Magnets attract objects which contain iron. The table gives examples of objects which contain iron and objects which do not contain iron.

Contains iron	Does not contain iron
key	eraser
nail	plastic spoon
needle	stick
paper clip	straw

In which group will all the objects be attracted to a magnet?

- (A) key, needle, plastic spoon, stick
(B) eraser, plastic spoon, straw
(C) needle, paper clip, key
(D) nail, paper clip, straw

2. The graph shows the life span of some animals.



For how long do chimpanzees live?

- (A) 28 years (B) 35 years (C) 63 years (D) 85 years

3. The photograph shows a brush-tail possum.

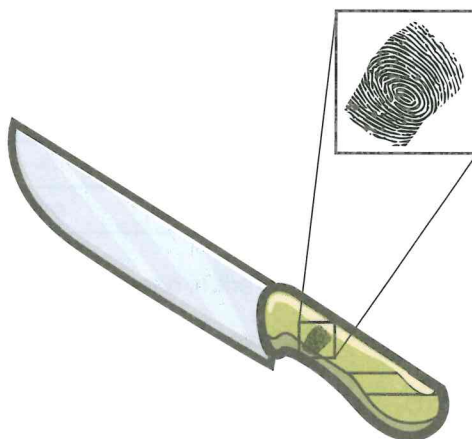


Which option does **not** describe the possum correctly?

- (A) The front paws have five clawed toes.
(B) It has large eyes for seeing at night.
(C) It has small rounded ears.
(D) It has a fur-covered body.

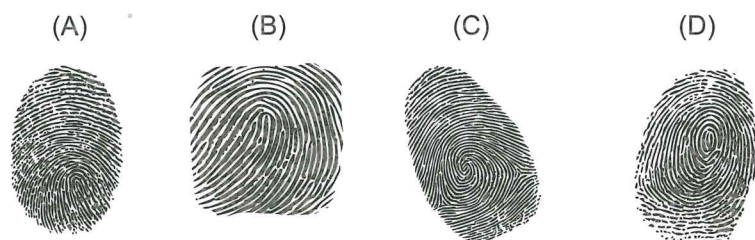
4. A fingerprint is an impression of ridge patterns on a person's finger. Each person has their own unique pattern.

Crime scene investigators found a partial fingerprint impression on a knife used at a crime.



The fingerprints of the suspects are shown below.

Which suspect used the knife?



5. Some whales feed on krill.

The picture shows the actual size of an average krill from the Southern Ocean.



What is the body length of the krill excluding the antennae?

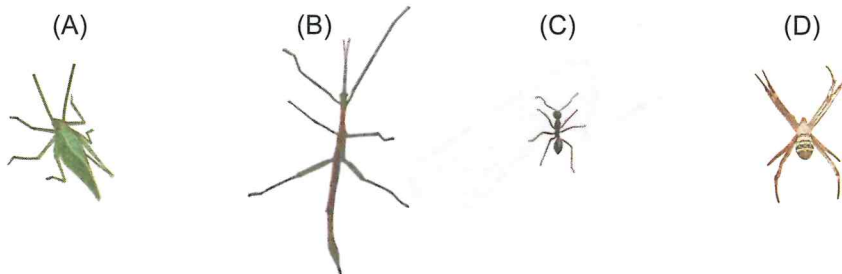
- (A) 14 mm
- (B) 29 mm
- (C) 35 mm
- (D) 45 mm

6. Insects are a group of small animals that have the following features:

- external skeletons
- three main body parts – head, thorax and abdomen
- three pairs of jointed legs
- one pair of antennae
- wings at some stage of their lives.

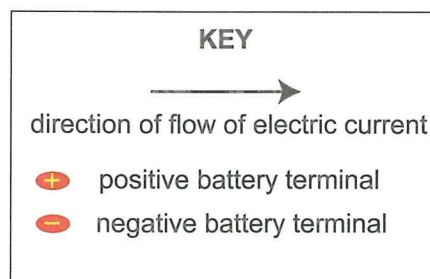
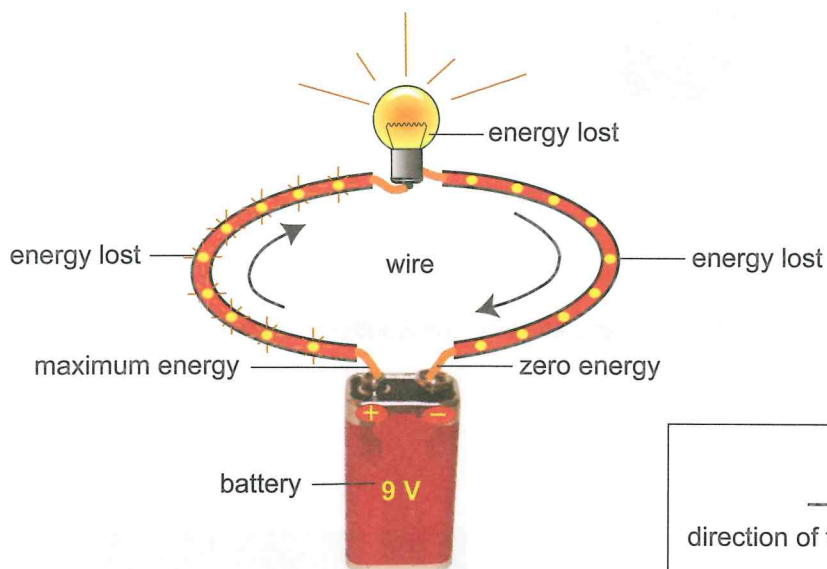
The photographs show four small animals.

Which animal is **not** an insect?

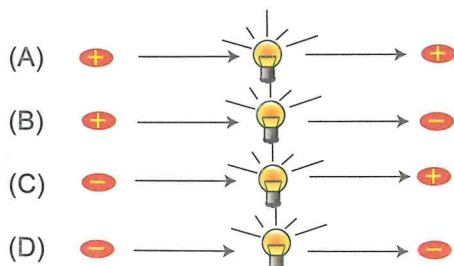


7. If a circuit is closed, electrical energy will pass through it.

The drawing shows a closed electrical circuit.

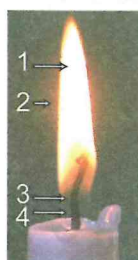


In which direction does the electric current flow in this circuit?



8. The photograph shows the zones that occur in the flame of a burning candle.

The table gives information about these zones.



Zone	Colour of light produced	Light is produced by
1	yellow	glowing particles of soot
2	red	fuel burning with oxygen
3	-	heat only, no oxygen present
4	blue	gassy fuel from candle

When fuel from a candle combines with oxygen in the air, it forms carbon dioxide and water.

What is the colour of the light produced by this process?

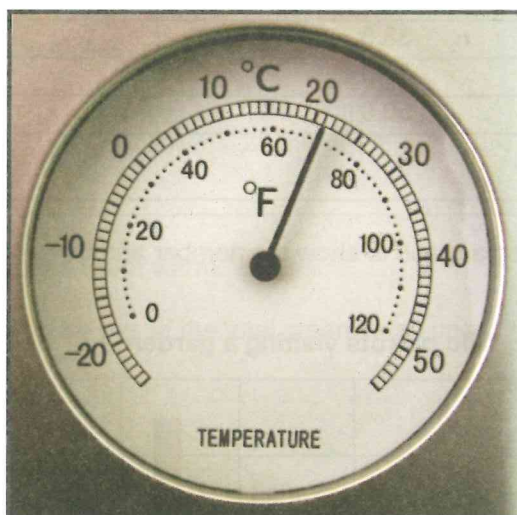
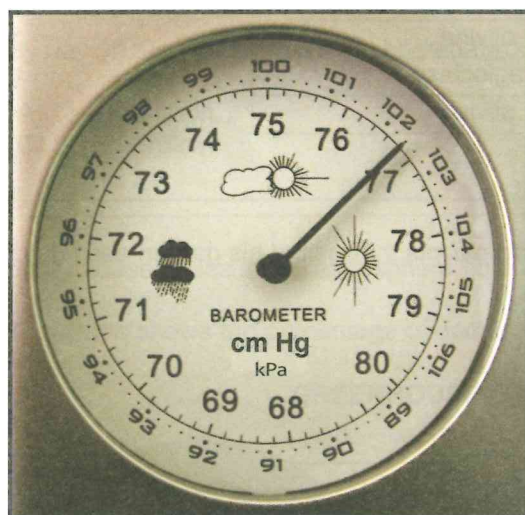
- (A) red (B) blue (C) colourless (D) yellow

For questions 9 and 10 use the information below.

Weather stations have a barometer and a thermometer. A barometer measures air pressure. A thermometer measures air temperature.

Temperature scales may be marked in degrees Celsius ($^{\circ}\text{C}$) or in degrees Fahrenheit ($^{\circ}\text{F}$).
Air pressure scales may be marked in centimetres of mercury (cm Hg) or in kilopascals (kPa).

The barometer and the thermometer shown in the photographs below have both scales.



9. What is the air pressure (kPa) and temperature ($^{\circ}\text{C}$) shown by the two instruments?

	Air pressure (kPa)	Temperature ($^{\circ}\text{C}$)
(A)	102.4	21
(B)	102.4	68
(C)	103.6	68
(D)	103.6	21

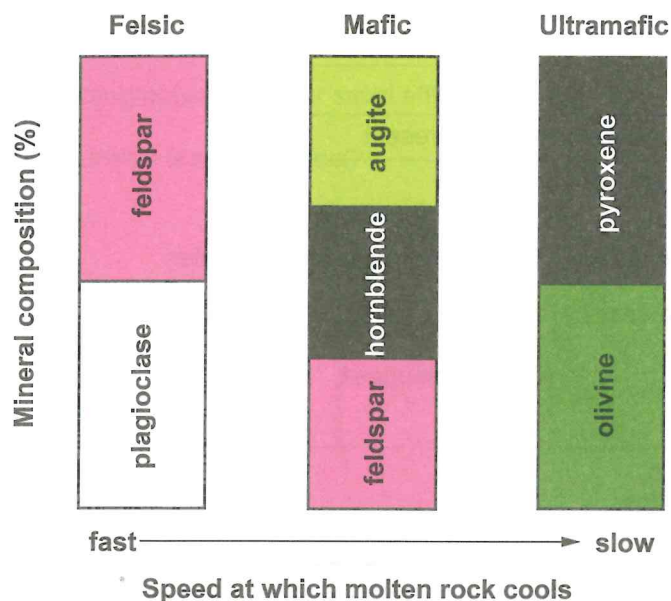
10. The symbols shown on the face of the barometer indicate the type of weather that might be expected at certain ranges of air pressure.

For what range of air pressures (kPa) would constant rain be most likely?

- (A) 94.1– 97.0
(B) 97.1–100.0
(C) 100.1–103.0
(D) 103.1–106.0

11. Depending on the rate at which it cools and its mineral composition, molten rock forms different types of rock: felsic, mafic or ultramafic.

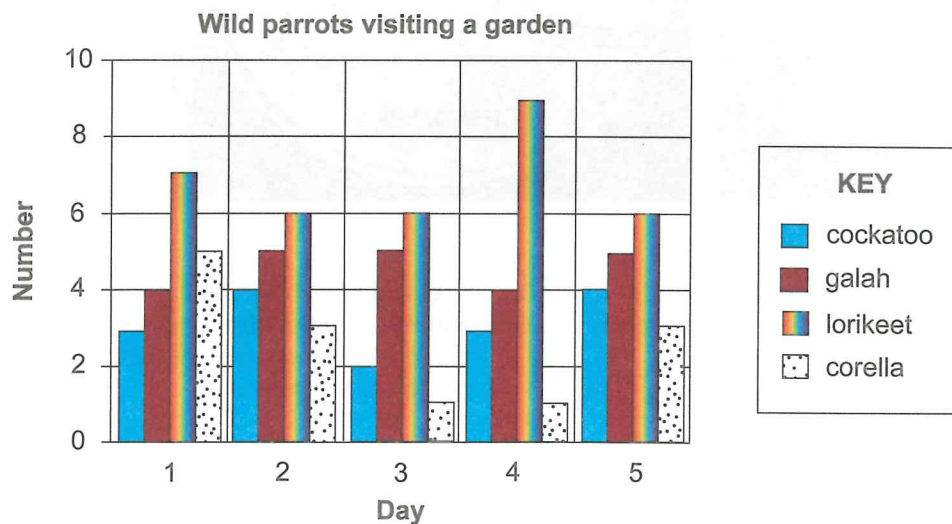
The graphs give information about minerals that make up these three rock types and the speed at which the rocks cool.



What proportion of felsic rock is made up of feldspar mineral, and which mineral is most likely to be formed in molten rock that cooled very slowly?

	Proportion of felsic rock (%)	Mineral formed
(A)	33.3	plagioclase
(B)	33.3	olivine
(C)	50.0	plagioclase
(D)	50.0	olivine

12. Sam constructed a graph to show the number and types of wild parrots that visited his garden each day for five days.



If only one wild parrot visited his garden on the sixth day, which type of parrot is this most likely to be?

- (A) cockatoo (B) galah (C) lorikeet (D) corella

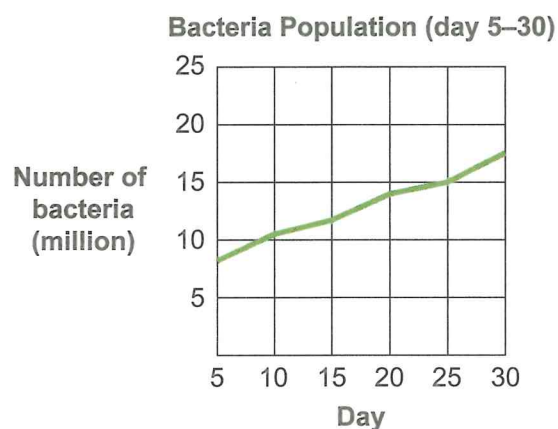
13. The table gives information about substances that skin secretes to form an acid mantle layer.

Substance type	Secreted from	Function of substance
sebum (oil)	hair follicles	protects skin and hair from wind and sunlight
sweat (slightly acidic, watery solution pH 4–4.5)	skin pores	slows growth of harmful bacteria

What protects the skin by slowing the growth of harmful bacteria?

- (A) hair follicles (B) sebum (C) skin pores (D) sweat

14. The graph shows the number of bacteria in a sample over a period of time.

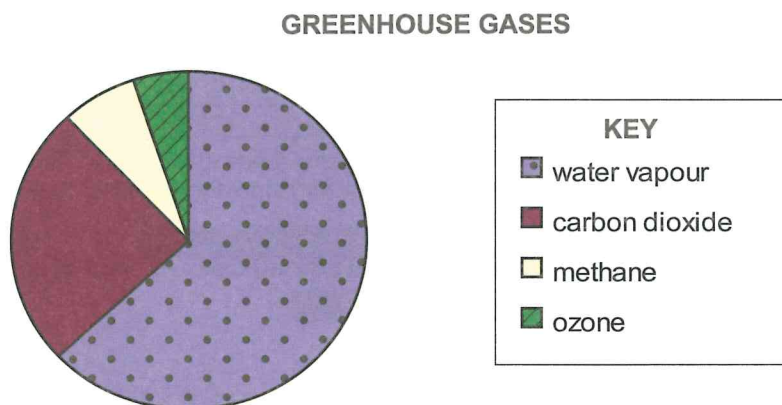


By day 35 the population of the bacteria would be closest to

- (A) 15 million. (B) 17 million.
(C) 20 million. (D) 25 million.

15. Greenhouse gases in the atmosphere help keep the Earth warm.

The graph shows the percentage of each greenhouse gas of the total greenhouse gases.



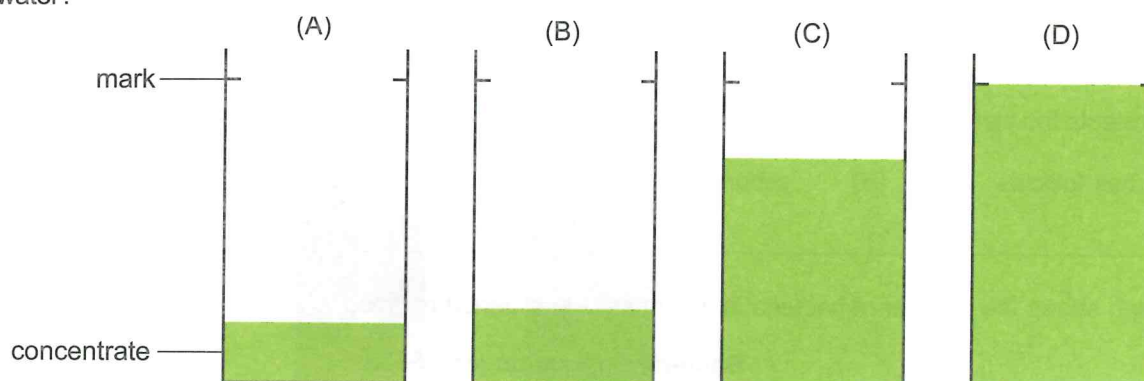
Approximately what percentage of total greenhouse gases is carbon dioxide?

- (A) 10% (B) 25% (C) 60% (D) 75%

16. Jenny bought a bottle of fruit juice concentrate. The instructions said that she should mix 1 part concentrate to 4 parts water.

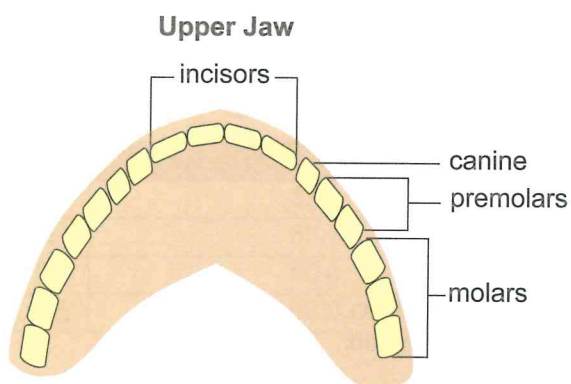
She wanted to make one container of fruit juice.

Which container shows the correct amount of concentrate that she should use, before filling to the mark with water?



17. Most mammals have four types of teeth: incisors, canines, premolars and molars.

The diagram shows the teeth in an adult human upper jaw.



A dental formula identifies the type and number of teeth in one half of each jaw in the following order: incisors, canines, premolars and molars.

Humans have the same type and number of teeth in the top and bottom jaws.

Which option shows the dental formula for an adult human?

- (A) 2.1.2.3 (B) 2.2.2.3 (C) 4.2.4.6 (D) 8.4.8.12

18. Khalid wanted to find out which of the four laundry liquids he had was better for removing cooking oil stains. He carried out the following steps:

- He dipped a large piece of cloth into cooking oil and then cut the oil-stained cloth into four equal pieces.
- He mixed each laundry liquid with warm water.
- He used each mixture to scrub a piece of oil-stained cloth.

What conditions should Khalid keep the same to have a fair test?

- (A) type of laundry liquid, type of oil, amount of stain left
(B) water temperature, amount of laundry liquid, scrubbing time
(C) amount of laundry liquid, scrubbing time, amount of stain left
(D) water temperature, amount of laundry liquid, type of laundry liquid

19. Leaves have veins for transporting water.

Figure 1 shows the vein structure of leaves from two different types of flowering plants.

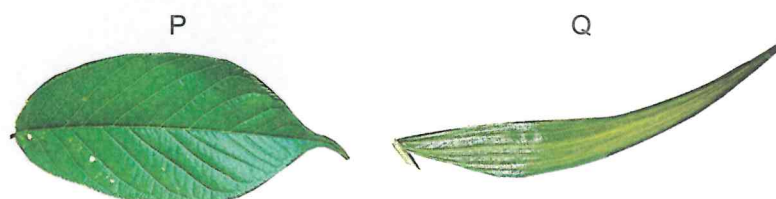


Figure 1

Leaf P will most likely be found on plants that have flowers with multiples of 4 or 5 petals.

Leaf Q will most likely be found on plants that have flowers with multiples of 3 petals.

Figure 2 shows the flowers from four different plants.

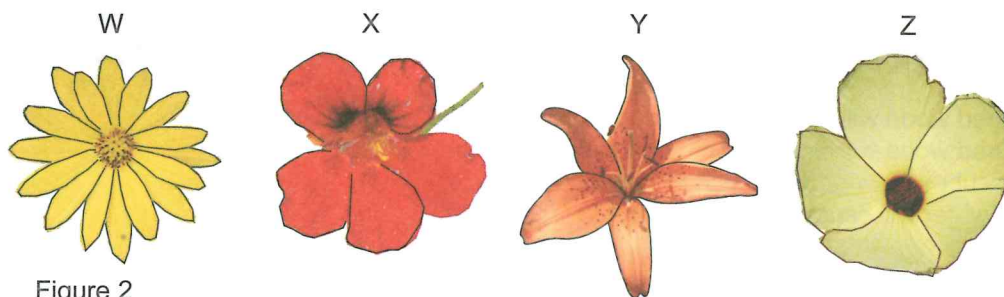


Figure 2

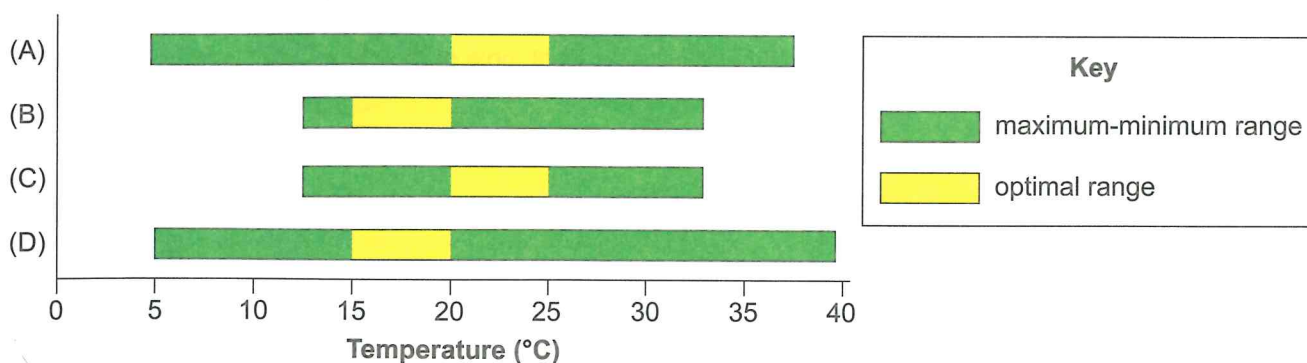
Which flower(s) will most likely be found on a plant with leaf type Q?

- (A) X and Z only
- (B) W and Y only
- (C) Z only
- (D) Y only

20. The table gives information about the temperatures at which crops will grow.

Crop	Minimum temperature (°C)	Optimal temperature (°C)	Maximum temperature (°C)
corn	20	22–25	34
potato	12	15–20	34
rice	18	30–33	40
wheat	5	20–25	38

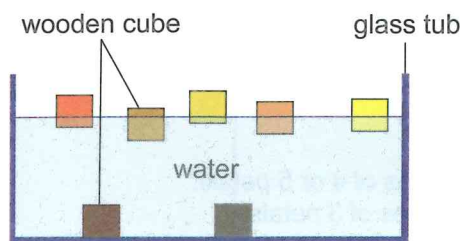
Which graph best represents potato?



21. Fran predicted that all types of wood will float on water.

She set up an experiment to test her idea. She collected cubes of the same size that were made of different types of wood. She placed them into a glass tub filled with water.

The diagram shows her results.



Fran concluded that her first prediction was incorrect.

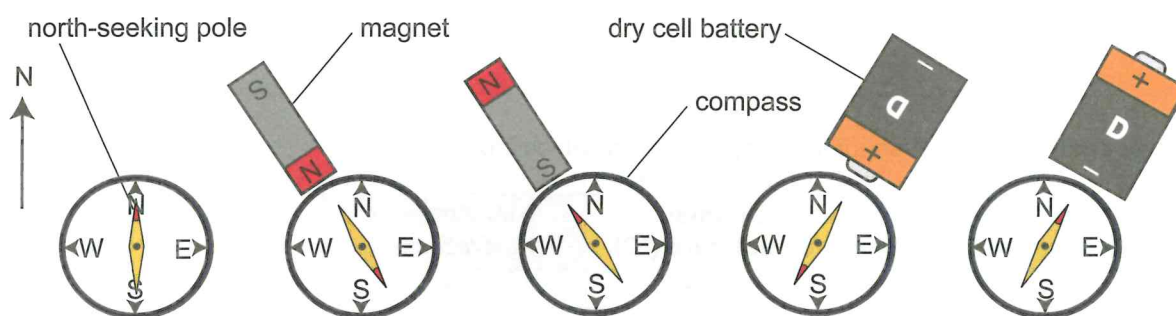
To continue her investigation, what change should Fran make to her prediction given her results so far?

- (A) Lighter coloured wood will float on water, dark coloured wood will sink.
- (B) Lighter coloured wood will sink in water, dark coloured wood will float.
- (C) Using bigger cubes of wood will cause more wood to sink.
- (D) Adding more water will cause all wood to float.

22. Goran wanted to investigate this idea: for a magnet, like poles repel and unlike poles attract.

He used a magnetic compass, a bar magnet and a type D dry cell battery for his investigation.

The diagrams show Goran's results.



Goran made four observations during his investigation. Which observation was **INCORRECT**?

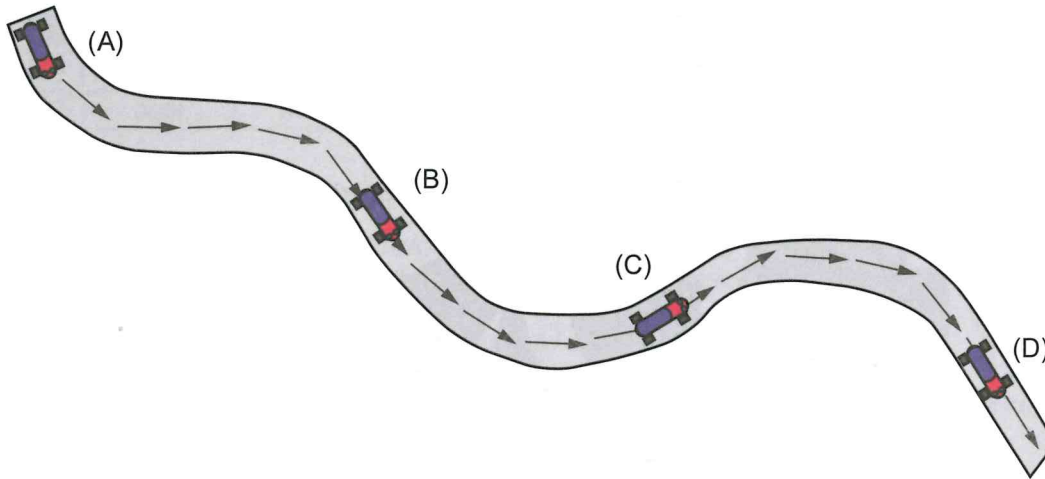
- (A) The north pole of the bar magnet repelled the north-seeking pole of the compass needle.
- (B) The negative end of the battery repelled the north-seeking pole of the compass needle.
- (C) The north-seeking pole of the compass needle pointed to the north when no magnet or battery was present.
- (D) The positive end of the battery behaved like the north pole of the bar magnet when placed near the compass.

23. The photograph shows a driver's view through her front windscreen at a particular point during a journey.

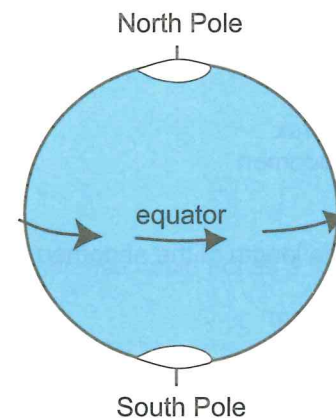


The map below shows the driver's journey from (A) to (D).

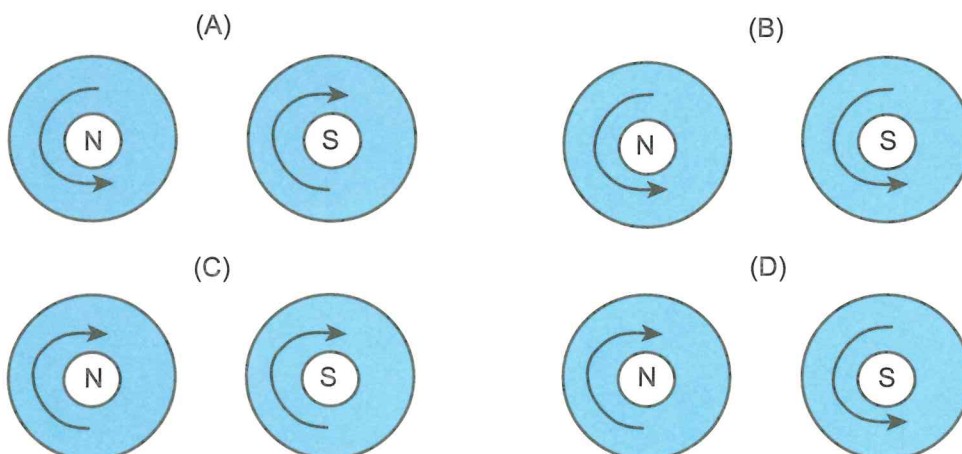
At which point along the journey would the driver see the view shown in the photograph?



24. The drawing shows how the Earth rotates when looking from above the equator.

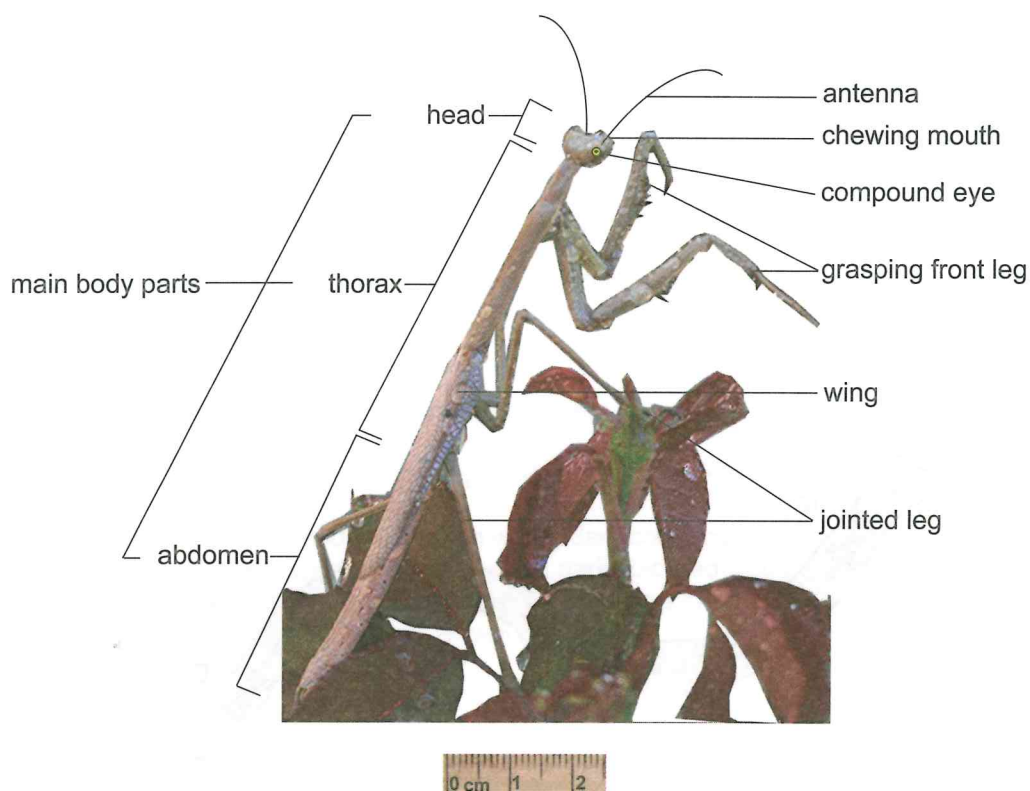


Which two drawings show how the Earth rotates when looking from above the North Pole and when looking from above the South Pole?



For questions 25 and 26 use the information below.

The scaled photograph shows external body parts of an adult praying mantis.



25. To which body part are the legs attached?

- (A) wing
- (B) head
- (C) thorax
- (D) abdomen

26. What is the length of the abdomen of this mantis?

- (A) 3.6 cm
- (B) 4.2 cm
- (C) 5.0 cm
- (D) 8.0 cm

27. Shanta considered cockroaches a pest. She read on a website that cockroaches prefer particular environmental conditions of temperature, moisture and light.

She set up eight environments to investigate which conditions attract the greatest number of cockroaches.



cool dry light	cool dry dark	X	cool moist dark
warm dry light	Y	warm moist dark	warm moist light

To make a fair test, what conditions did Shanta need to set for environments X and Y?

	Conditions for environment X			Conditions for environment Y		
(A)	cool	dry	light	warm	dry	light
(B)	cool	dry	dark	warm	moist	light
(C)	cool	moist	dark	warm	moist	dark
(D)	cool	moist	light	warm	dry	dark

28. Solids have a fixed volume and hold their shape.
Liquids have a fixed volume and take the shape of any container into which they are placed.

Liquids and gases are fluids. Fluids spill or flow away if they are not contained.

The diagram shows four containers of different substances. The substances were all kept in a freezer at a temperature which was below 0 °C.



butter wood sand ice

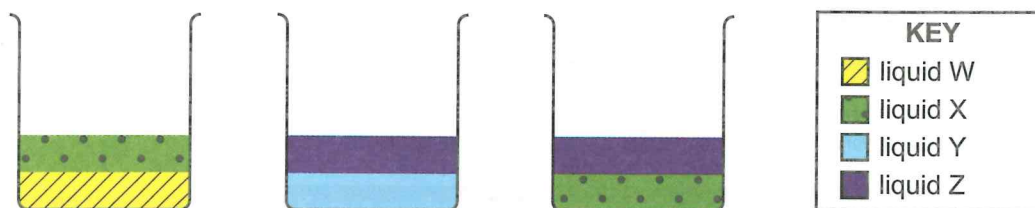
If the containers were removed immediately from the freezer, which substance could act as a fluid?

- (A) butter
- (B) wood
- (C) sand
- (D) ice

29. Misha has four different liquids, W, X, Y and Z.

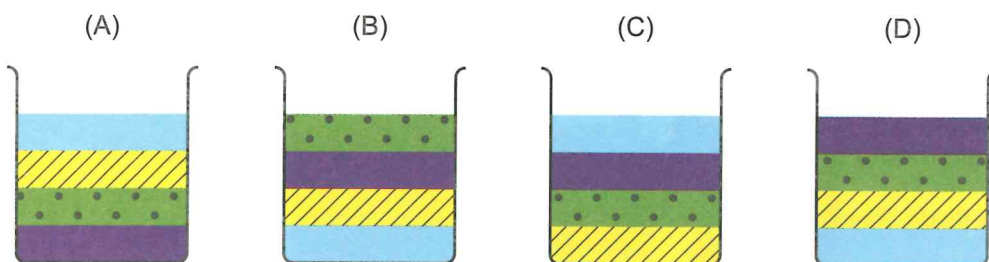
When the liquids are poured together they do not mix but form separate layers.

The diagrams show what happened when Misha made three different combinations from these liquids.



Afterwards, Misha combined all four different liquids.

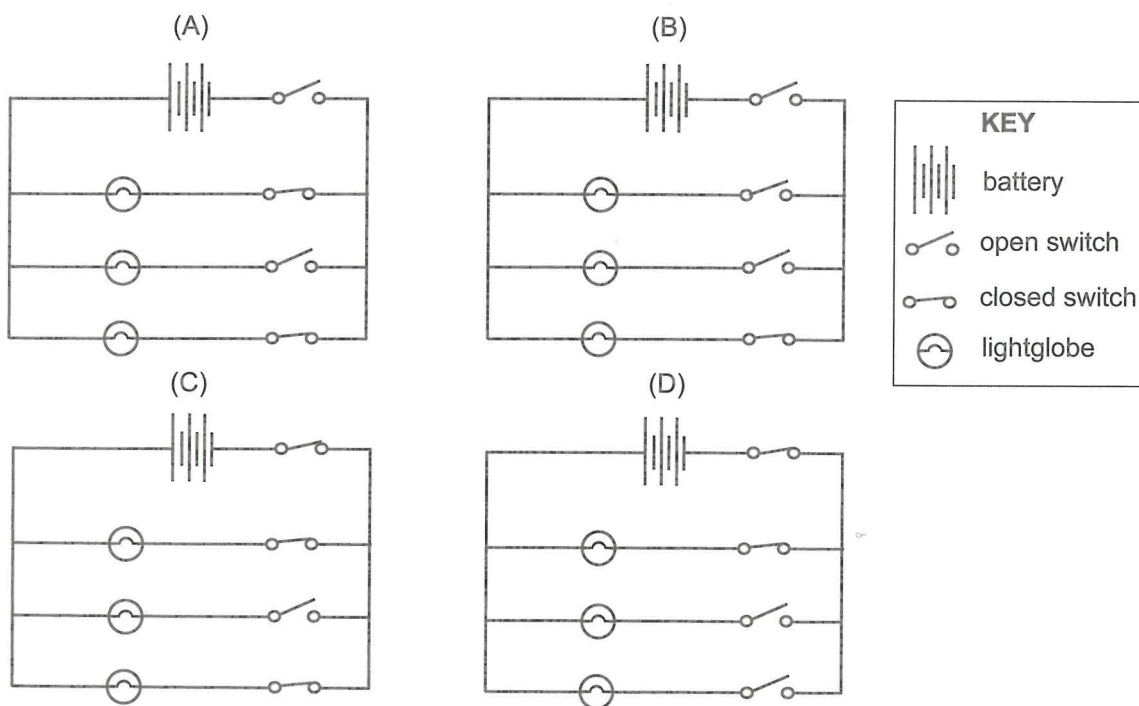
Which of the following could show Misha's results?



30. A circuit diagram shows how an electrical circuit is joined together. Standard symbols are used in circuit diagrams to represent electrical components.

Electrical current will flow only when a circuit is closed. Lightglobes will glow only if electrical current is flowing.

The diagrams show four circuit diagrams. In which circuit would only one lightglobe glow?

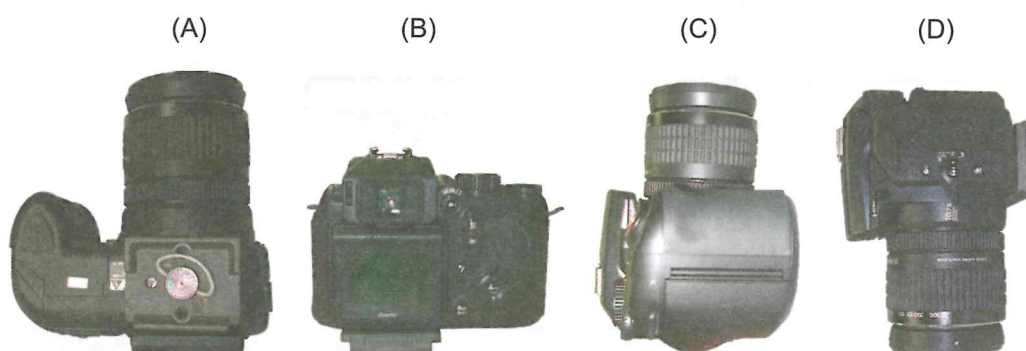


31. The photographs show views of Jan's camera from the front and from the top.



Jan rotated the camera and photographed it from four directions. The photographs are shown below.

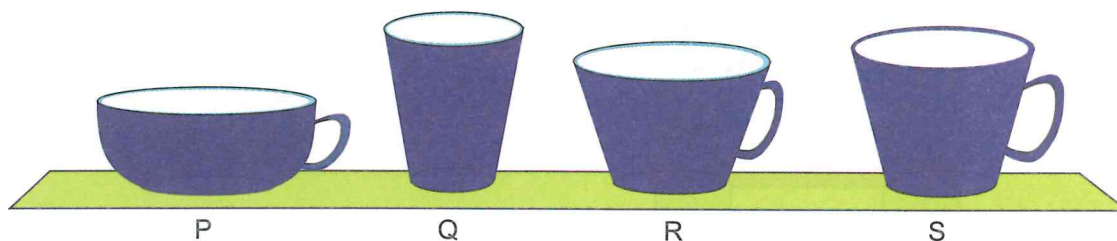
Which photograph shows the view from the direction given by the arrow?



32. Simon investigated how quickly heat escaped from cups that are made of the same insulating material.

He filled four 250 mL cups with boiling water. He measured the temperature of the water in each cup with a thermometer every 2 minutes. He stopped his measurements when one of the cups reached 35 °C.

The diagram shows the cups he used.

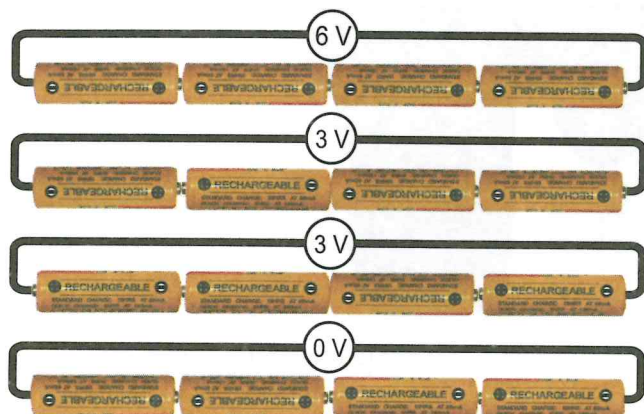


Which option is a correct conclusion that Simon could make from this investigation?

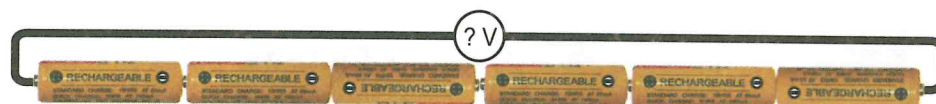
- (A) It was better to use cup R than cup S if you wanted to keep water hot.
(B) The water in cup Q reached 35 °C before the water in cup S.
(C) Cup P was the worst cup for keeping water hot.
(D) The water in cup S was the first to reach 35 °C.

33. Sean investigated the total voltage produced in a circuit by different arrangements of dry cell 1.5 volt (V) batteries.

Sean's results are shown below.



Sean increased the number of batteries in the circuit.

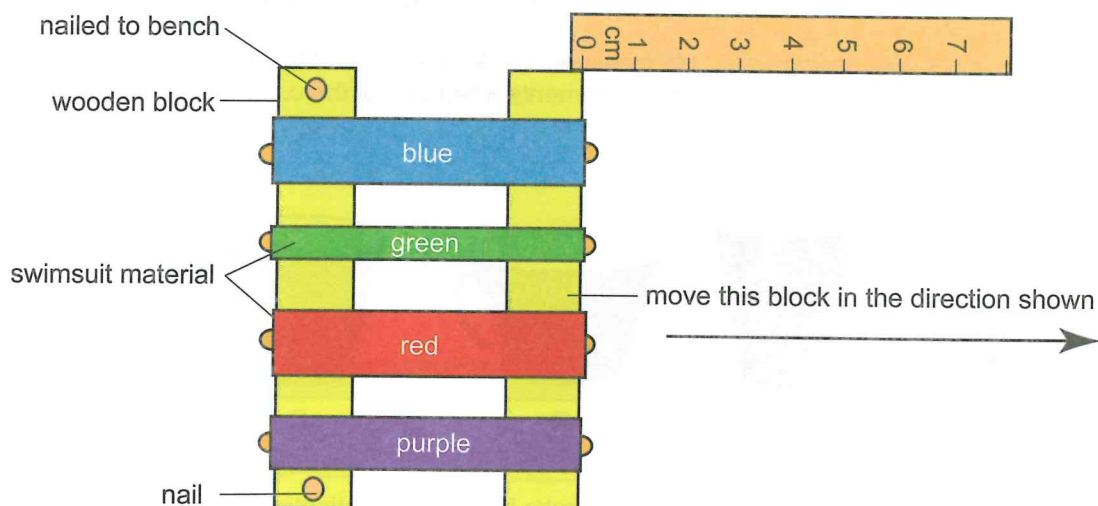


What voltage should this new arrangement produce?

- (A) 0 V (B) 3 V (C) 6 V (D) 9 V

34. Jess investigated the stretchability of four different swimsuit materials to see which material would stretch the furthest.

She set up her investigation as shown.



Jess' teacher advised her that she had a problem with her testing method that would affect the fairness of her investigation.

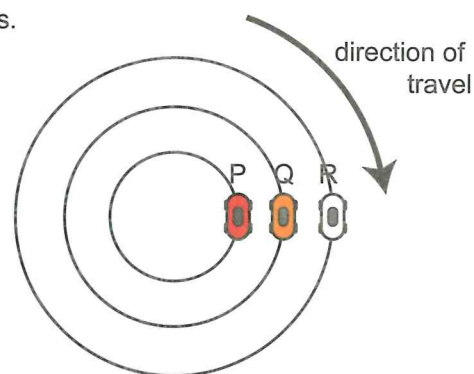
Which modification to her method would improve the fairness of Jess' investigation?

- (A) Nail the second wooden block to the bench.
 (B) Make the four strips of material the same colour.
 (C) Make the four strips of material the same width.
 (D) Double the number of nails holding the wider strips of material.

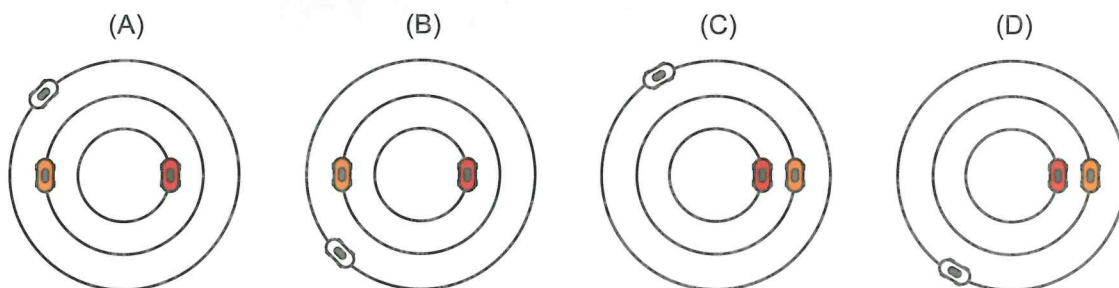
35. Three toy trains P, Q and R are travelling around three separate tracks. All three trains are travelling at the same speed.

Train P takes 1 minute to completely travel around its track.
Train Q takes 2 minutes to completely travel around its track.
Train R takes 3 minutes to completely travel around its track.

At certain times the three trains are level with each other as shown.

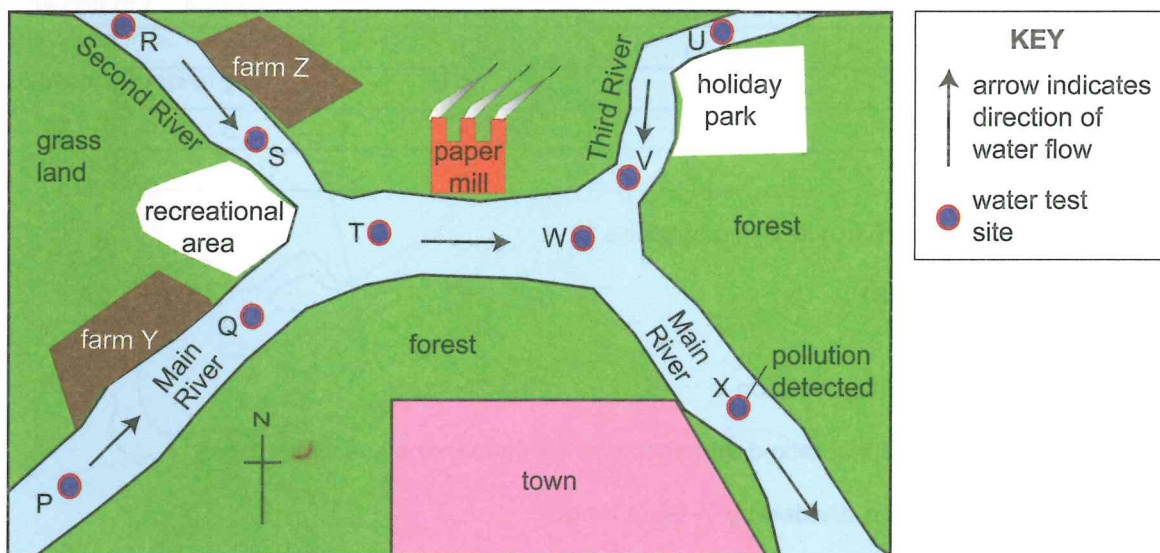


Which diagram shows the position of the trains five minutes later?



36. Pollution in a river can be detected by testing the water from the river.

The map shows the land use around a river and the location of several water testing sites (P – X).



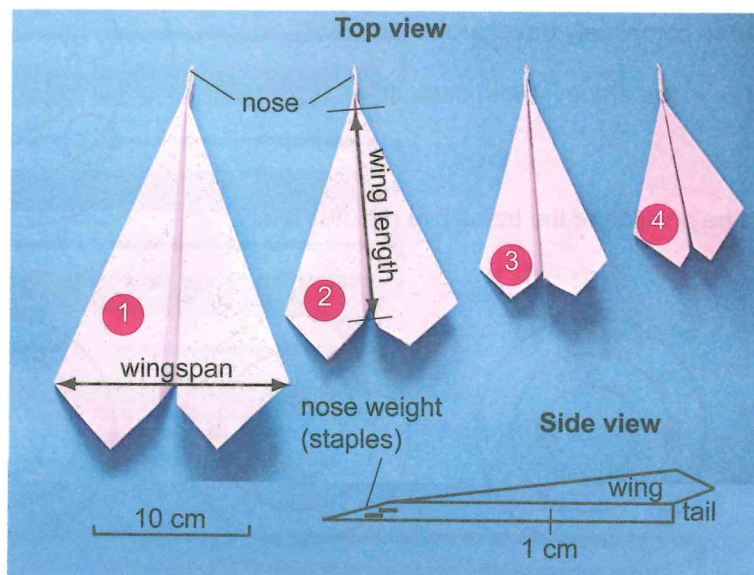
Pollution was first detected in water samples at test site X. Scientists investigated to find the source of the pollution. They concluded farm Y was the source of the pollution.

For the scientists to reach this conclusion, at which test sites was pollution detected, and which test sites were clear of pollution?

	Sites where pollution was detected	Sites that were clear of pollution
(A)	P, Q, T, W	R, S, U, V
(B)	P, R, S, U, V	Q, T, W
(C)	Q, T, V, W	P, R, S, U
(D)	Q, T, W	P, R, S, U, V

For questions 37 and 38 use the information below.

Ben constructed four paper planes. He predicted that the plane with the biggest wingspan would glide best. The photograph shows the four planes to scale and a side view of their construction.



Ben trialled his planes in a large room by throwing each plane in the same way. He measured the distance each plane glided from where it was thrown. His results are shown below.

Distance travelled by each plane (m)				
Trial	Plane 1	Plane 2	Plane 3	Plane 4
1	6	5	3	8
2	5	5	3	7
3	7	4	3	8
4	6	5	4	7

37. What was the average distance (m) travelled by plane 1?

- (A) 5 (B) 6 (C) 7 (D) 24

38. Ben's results did **not** support his prediction.

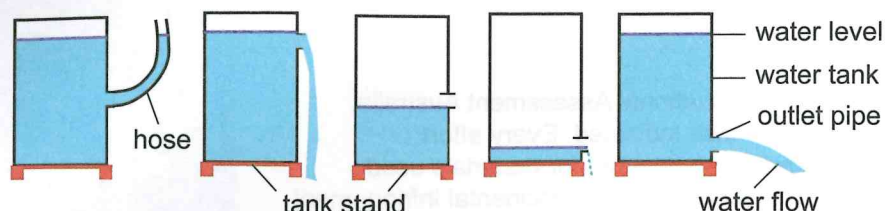
He discovered that a wing's aspect ratio determines how far the paper plane would travel.

aspect ratio = wingspan divided by $\frac{1}{2}$ wing length

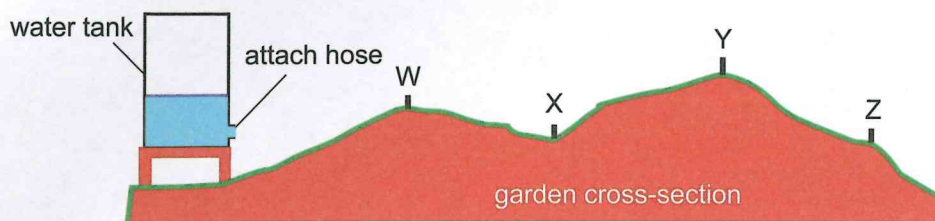
What is the aspect ratio of plane 4?

- (A) 1.5 (B) 2 (C) 7 (D) 9.6

39. The diagram shows how the water flow from a cylindrical water tank is affected by the water level in the tank and the position of the outlet pipe.



Jan placed a cylindrical water tank in her garden. The diagram shows the level of the water in her tank and the heights and locations of points W, X, Y and Z in her garden.



Jan attached a plastic hose to the outlet pipe in the position indicated. The hose was long enough to reach all points in the garden.

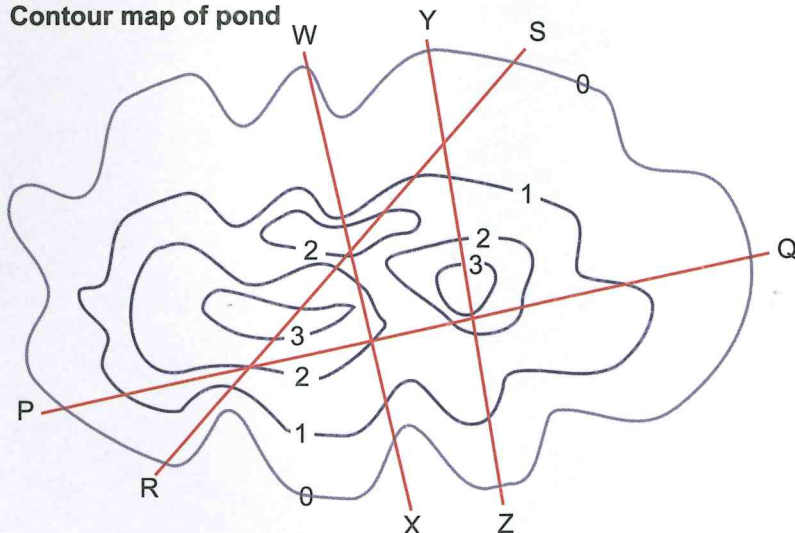
With the conditions shown, which points in her garden could Jan water using the hose?

- (A) W, X, Y and Z (B) W, X and Z only (C) W and X only (D) Y only

40. Contour lines on a map join points of equal height or depth.

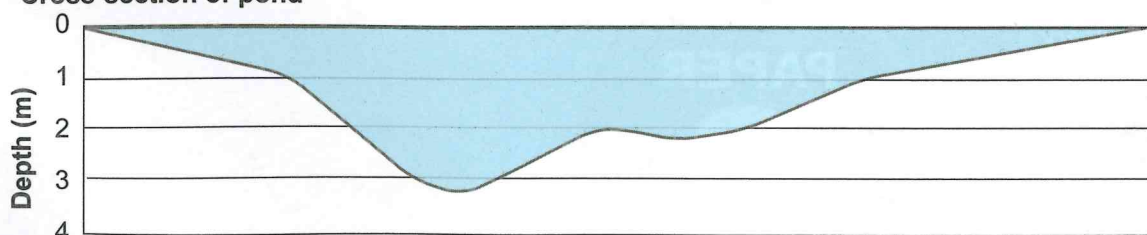
The contour map shows the depths (m) in a pond. There are four cross-sectional lines drawn across the map: PQ, RS, WX and YZ.

Contour map of pond



The graph below shows the water depth along one of the cross-sections.

Cross-section of pond



Which cross-section is shown above?

- (A) P to Q (B) W to X (C) Y to Z (D) R to S

ACKNOWLEDGMENT

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**PAPER
C**

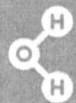


SC | 2009 Science Answer Keys

ICAS
 International Competitions
 and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Paper I	Paper J
1	D	A	C	A	C	A	A	B	D	D
2	D	A	B	C	B	B	A	A	D	D
3	C	C	C	A	A	A	C	C	B	B
4	C	D	D	B	D	A	C	B	D	D
5	D	B	B	B	A	D	C	D	C	C
6	A	C	D	B	B	A	C	B	C	C
7	B	A	B	C	D	C	A	C	B	B
8	D	B	A	D	B	B	C	B	B	B
9	C	A	A	B	A	D	A	B	B	B
10	A	D	A	A	B	B	B	B	B	B
11	A	C	D	D	D	A	B	C	C	C
12	C	B	C	A	A	A	D	A	D	D
13	D	B	D	C	B	C	D	C	A	A
14	C	D	C	C	A	C	B	D	D	D
15	C	A	B	D	A	B	B	B	C	C
16	B	C	A	B	C	C	B	D	D	D
17	A	A	A	D	C	D	D	A	C	C
18	B	D	B	D	A	D	B	A	C	C
19	B	D	D	B	C	D	A	A	A	A
20	A	B	B	A	D	C	A	B	A	A
21	D	D	A	D	D	B	C	B	C	C
22	D	B	B	B	C	C	B	C	A	A
23	D	A	C	C	C	D	C	C	A	A
24	C	C	A	D	C	C	C	A	B	B
25	B	C	C	C	C	B	A	C	A	A
26	A	C	B	B	C	A	B	A	A	A
27	A	D	D	D	D	C	C	D	D	D
28	B	D	C	C	A	C	D	C	C	C
29	C	B	D	C	D	B	B	A	B	B
30	B	A	D	B	B	B	D	B	A	A

(Please turn over)



SC

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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Paper I	Paper J
31			D	B	A	A	C	D	B	B
32			C	C	C	D	A	D	B	B
33			B	A	D	B	B	D	B	B
34			C	D	D	D	B	B	D	D
35			A	B	B	D	A	D	C	C
36			D	A	B	A	B	A	D	D
37			B	A	B	B	B	D	D	D
38			A	D	B	C	D	C	D	D
39			C	A	D	D	D	D	C	C
40			D	B	D	D	A	A	B	B
41					C	C	D	A	A	A
42					B	C	B	D	C	C
43					B	D	C	C	B	B
44					D	D	D	C	B	B
45					C	C	D	B	D	D

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